SIMATS ENGINEERING (SAVEETHA SCHOOLOFENGINEERING)

Department of Computer Science & Engineering

CO1 - ASSESSMENT TOOL -1

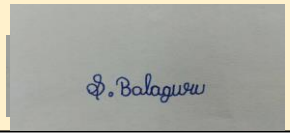
Descriptive Written Exam

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Descriptive Written Exam**
Weight age	**50 % of CIA**	**Total Marks**	**100**
CO1Statement	*Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.* **(BL3)**		
Student Name	**Balaguru S**	**Reg.No.**	**192312070**
Section/Batch	**Slot A / 2023**	**Date**	**21-10-2026**

Code of Conduct:

I, Balaguru S, certify that this submission is my original work and that I have adheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrity ruleswillresultindisciplinaryaction.

Signature of the student:



CSA3717 - Descriptive Written Exam CO1_AT-1

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3. A web application shows mismatch between expected and actual output during login and transaction operations. Apply basic testing definitions to identify errors, defects, and failures in this scenario. Illustrate their relationship with suitable examples. Explain how improper defect reporting clarity. Justify your answer with example.

Errors, Defects, and failures in a Web Application

1. Basic definitions:

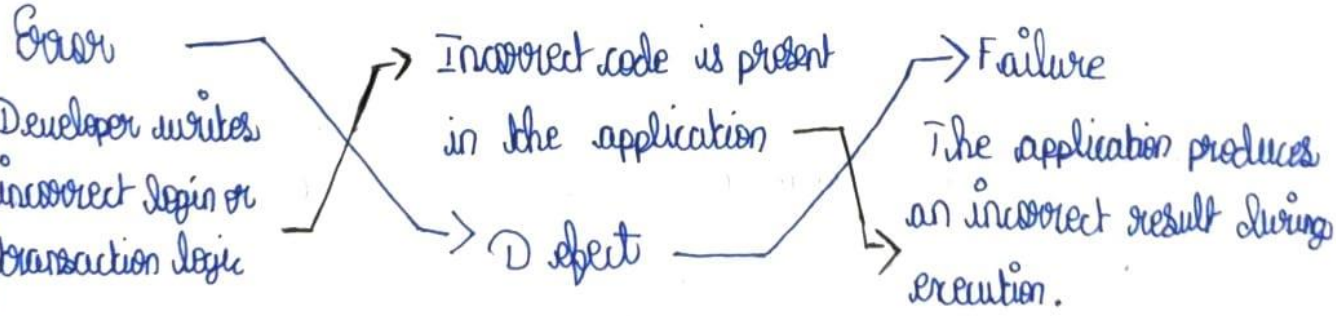
*) Errors : A human mistake made during software development, such as a Programmer writing an incorrect formula or misunderstanding a requirment.

*) Defects : A flaw in the software caused by an error, making the application behave incorrectly. (defect ≈ bug).

*) Failure : The observable incorrect behaviour of the application when the defect is executed.

2. Relationship between Error, Defect, and failure:

↳ The relationship can be illustrated as follow



3. Example in Login and Transaction operation:

Term	Login example	Transaction Example
Error	Developer misunderstands the validation requirement	Developer uses the wrong calculation formula.
Defect	Login code accepts an incorrect password.	Transaction code deducts ₹500 instead of ₹200.
Failure	An unauthorized user successfully logs in	The user's account is incorrectly debited by ₹500.

4. Effects of improper understanding:

*) Incorrect defect reporting: Testers may report a failure as a programming error without identifying the underlying defect.

*) Difficulty debugging: Developers may waste time searching the wrong code section.

*) Poor communication: Testers and developer may be treated as a miss understood each other's reports.

*) Incorrect prioritization: A serious transaction failure may be treated as minor visual issue.

*) Incomplete testing: Fixing the visible failure without addressing the root defect may allow the issue to occur again.

5. Improving defect reporting clarity:

↳ A report should include the following details

*) Defect ID:

*) Title:

*) Preconditions:

*) Steps to reproduce:

*) Expected result:

*) Actual result:

*) Severity & priority:

*) Evidence:

6. Justification:

↳ Understanding these terms helps testers distinguish the cause from the observable ~~failure~~ effect. Clear defect reports improves communication, speeds up debugging, support accurate severity assessment and help prevent similar problem through regression testing.

Errors made by people can introduce defects into software, and defects may cause failures during execution. However, not every error produces defect, and not every defect produces a failure in every situation.

Sample defect report :

DEF-001 : Incorrect debit amount during money transfer.

Precondition user is logged in and has sufficient balance.

Steps :

1. Open the transfer page.
2. Enter ₹200.
3. Confirm the transaction.

Expected :

Exactly ₹ should be deducted from the account

Actual

₹500 is deducted from the account

Severity

Critical or High, depending on the application's impact and classification.

7. Conclusion :

Correctly identifying errors, defects and failures, combined with clear and evidence-based defect reporting, improves software qu Security and accuracy.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

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